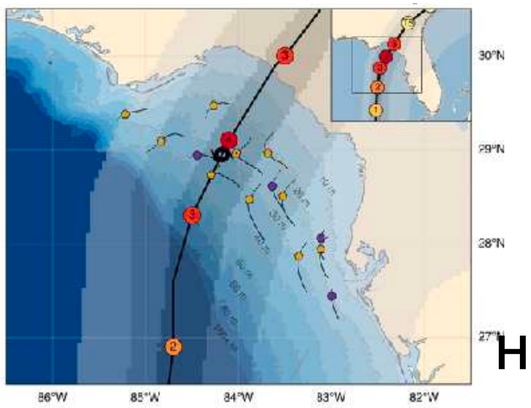
International Workshop on Tropical Cyclone–Ocean Interaction in the Northwest Pacific
Busan, South Korea, October 21-23, 2026

Directional Wave–Wind Coupling, Drag Coefficient Rolloff, and Air-Sea Fluxes Under Tropical Cyclones

Hyodae Seo & César Sauvage
Uehiro Center for the Advancement of Oceanography (UC•AO)
University of Hawai‘i at Mānoa

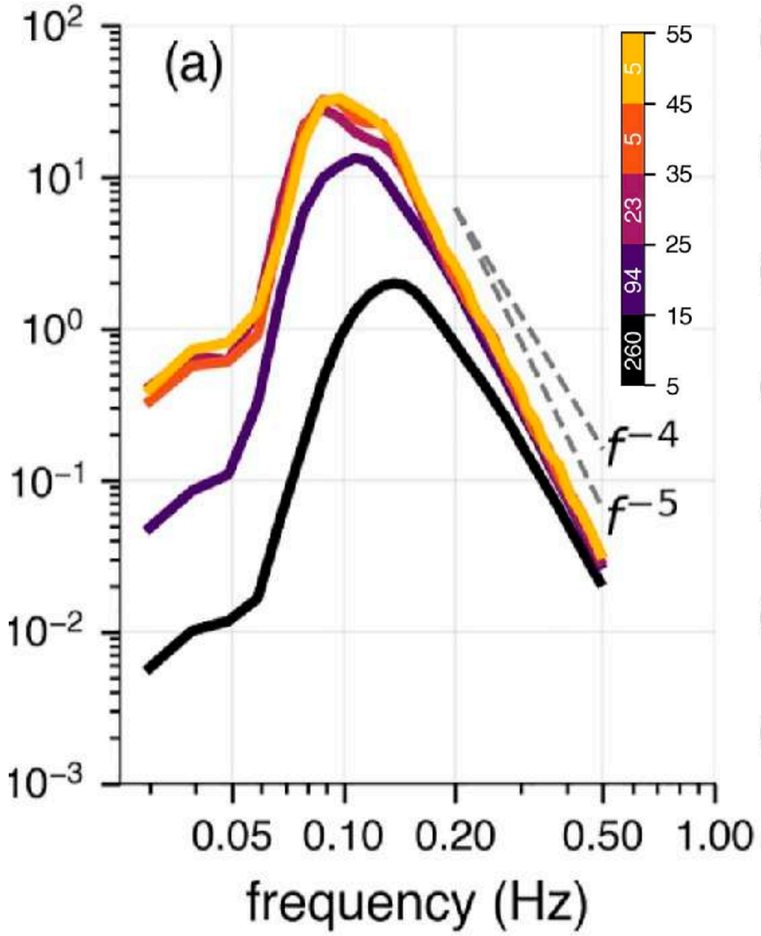
In collaboration with
Tunggul Bhirawa & Satoshi Mitarai
Okinawa Institute of Science and Techbnology

Saturation of wave energy and directional wind-wave

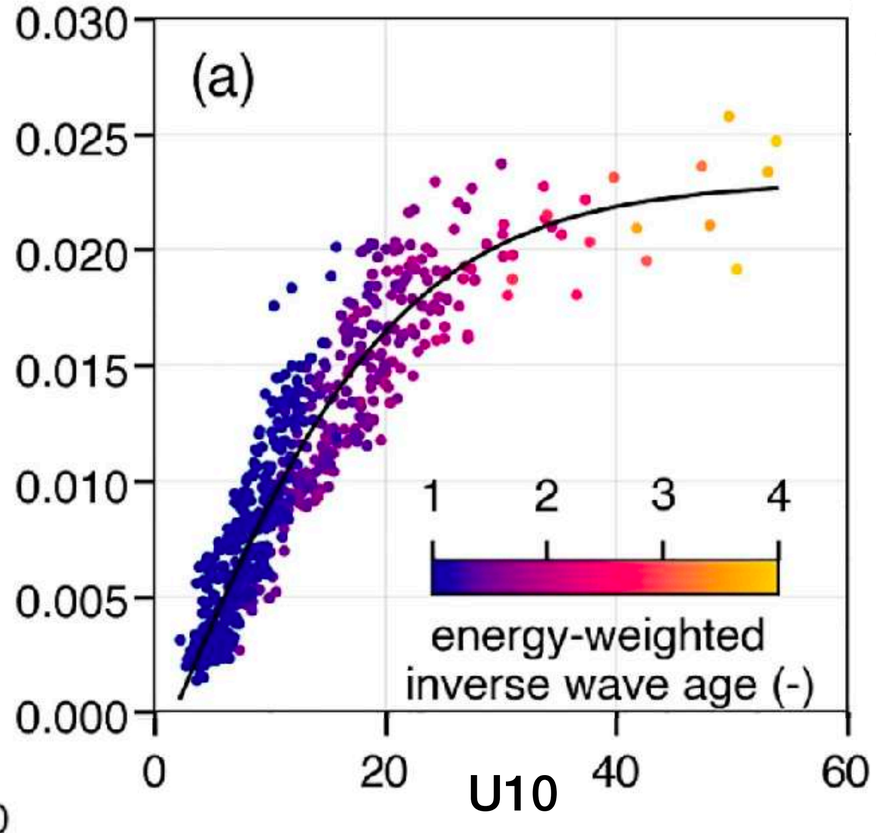


Hurricane Idalia (2023)

Mean energy density

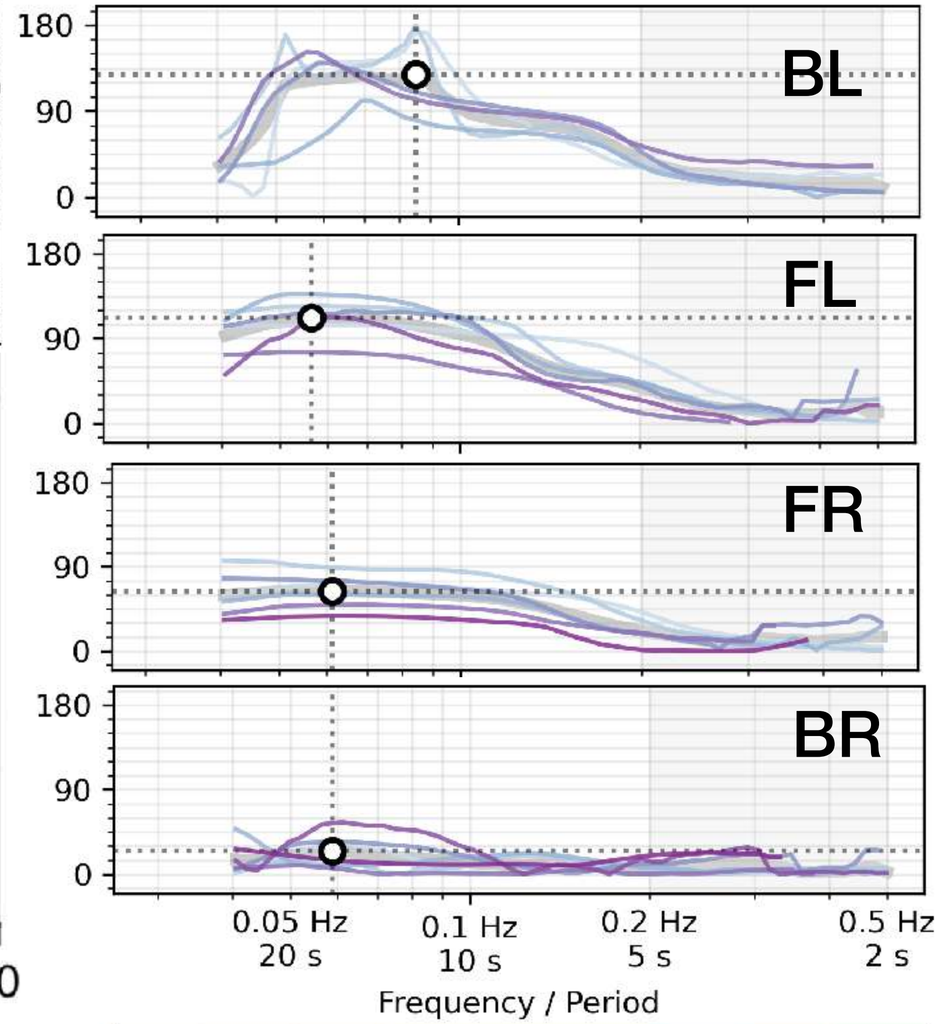


Mean square slope



Davis et al. (2025)

Directional wind-wave coupling



Pauly et al. (in prep)



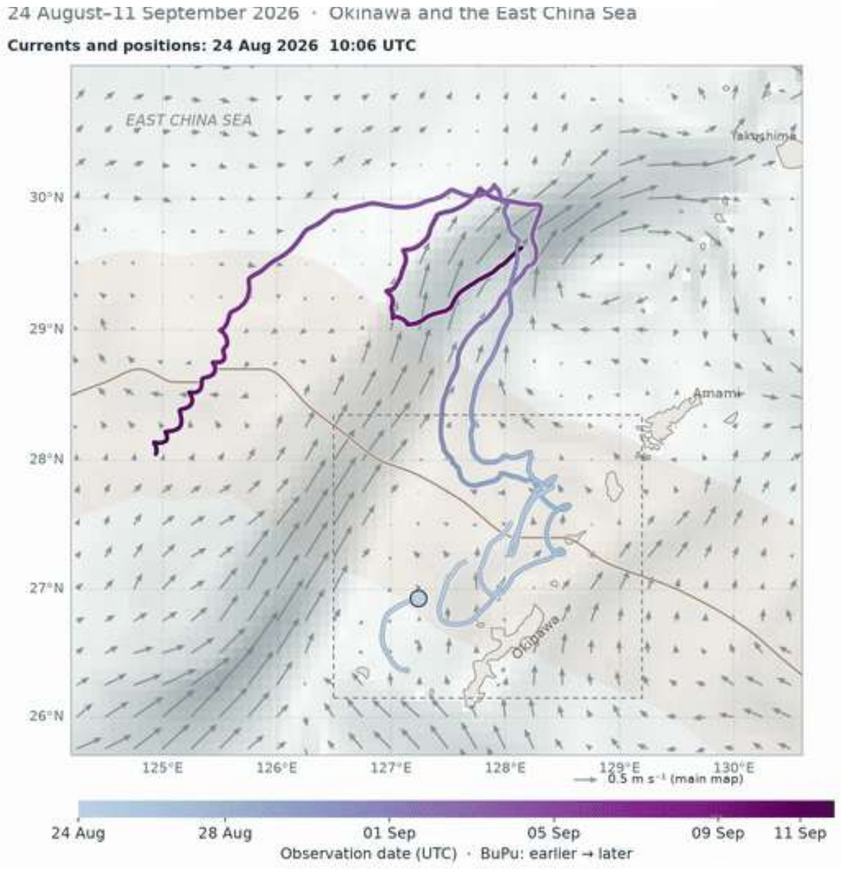
Wyn Pauly's poster: Typhoon Saudel (2026)

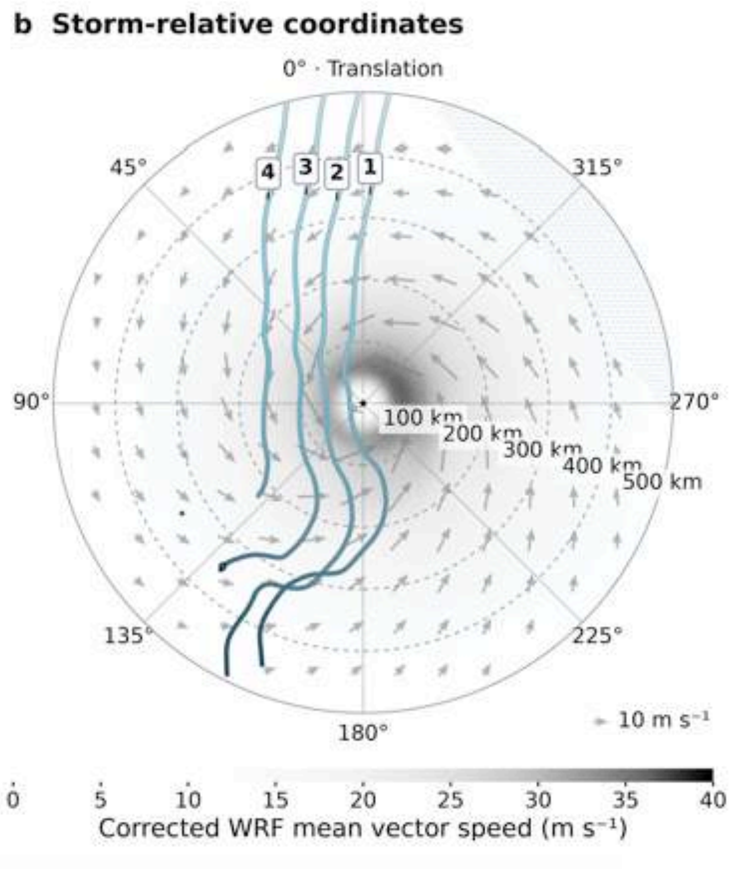
From Inherited Waves to Local Forcing:
Frequency-Dependent Wave Adjustment during Typhoon Saudel

How do we reconstruct directional wave evolution under a moving tropical cyclone?

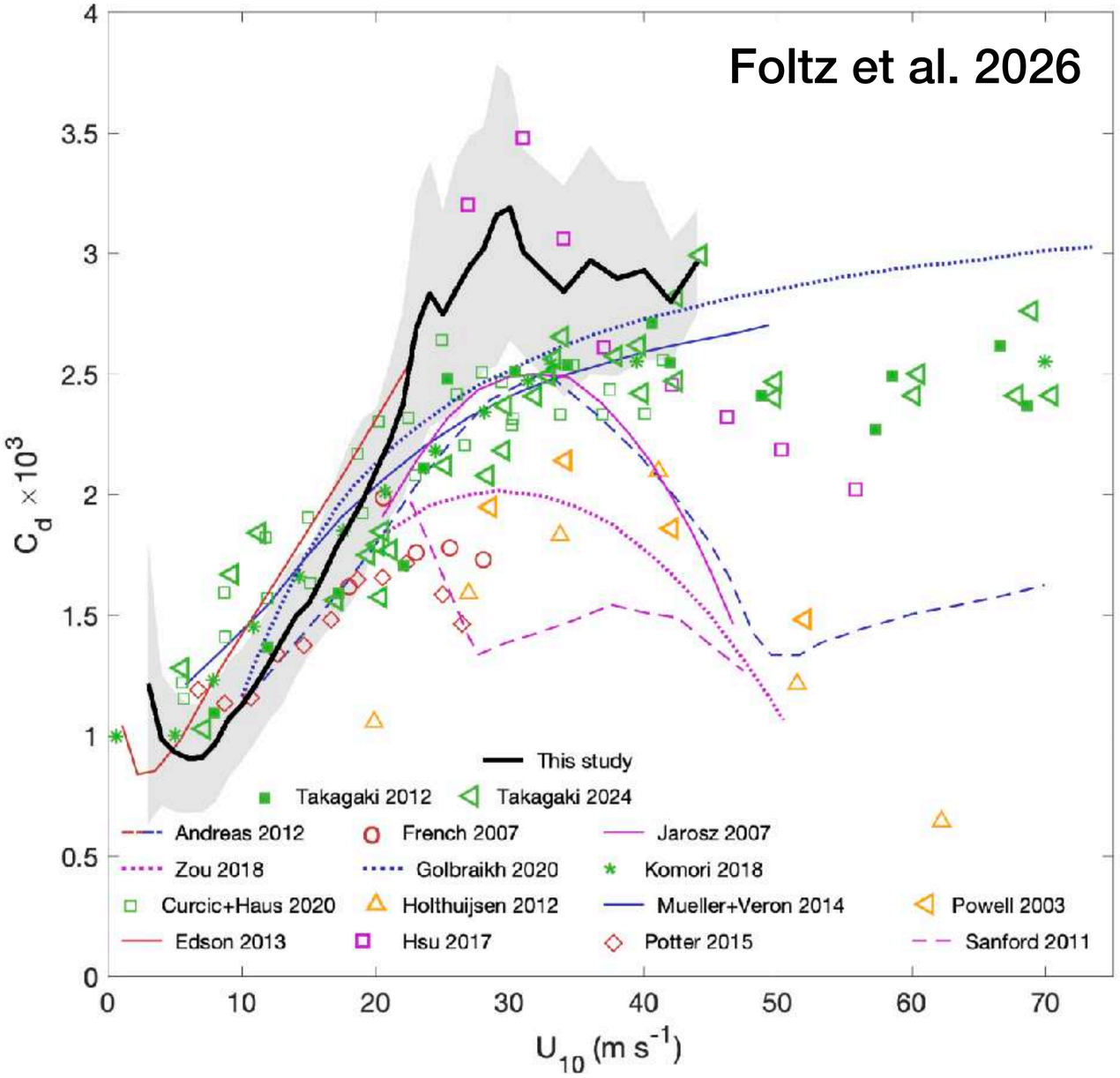
Four drifting Spotter buoys sampled the evolving wave field during Typhoon Saudel (24–28 August 2026).

Directional spectra are reconstructed from measured Fourier moments using the maximum entropy method (MEM), then corrected for Doppler shifts associated with buoy motion. The resulting spectra resolve frequency-dependent wave energy and directional spreading.





Drag coefficient (C_D) and roughness length (z_0) parameterizations



$$C_D = \left[\frac{\kappa}{\ln(z/z_0) - \psi_m(\zeta)}\right]^2$$

$$z_0^{\text{rough}} = \alpha \frac{u_*^2}{g}$$

$$z_o^{rough} = H_s 0.09 \left(\frac{u_*}{C_p}\right)^2$$

} Edson et al. (2013)

$$z_o^{rough} = H_s 0.09 cos(0.45\theta) \left(\frac{u_*}{C_p}\right)^{2cos(-0.32\theta)}$$

$$z_o^{rough} = H_s 0.09 \left(\frac{u_*}{C_m}\right)^2$$

} Sauvage et al. (2023)

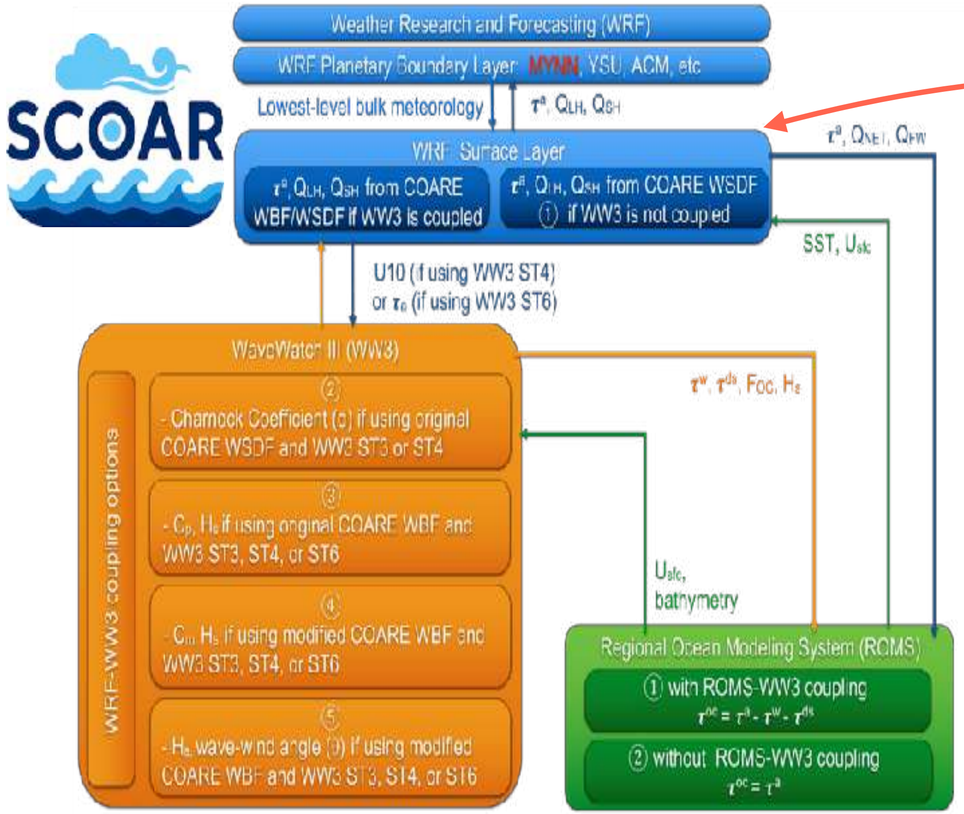
$$C_h = \left[\frac{\kappa}{\ln(z/z_0) - \psi_m(\zeta)}\right]\left[\frac{\kappa}{\ln(z/z_{0t}) - \psi_h(\zeta)}\right]$$

SCOAR WRF-ROMS-WW3 fully coupled simulations

using 3 different z_0 scenarios

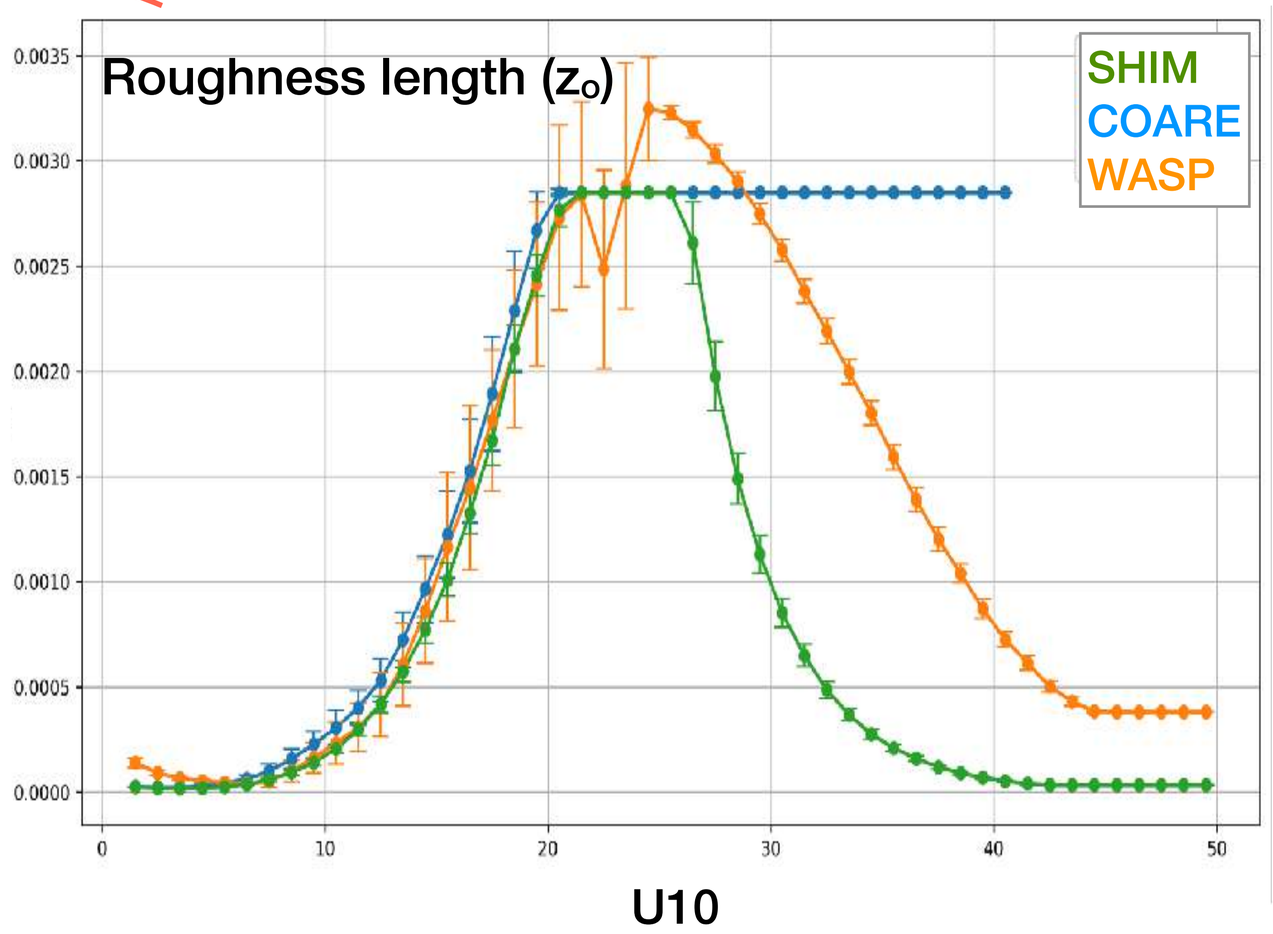
$$z_o^{rough} = H_s 0.09 \frac{u_*^2}{C_m}$$

- COARE (Edson et al. 2013; Sauvage et al. 2023)
- WASP (Bouin et al. 2024)
- SHIM (Shimura et al. 2024)

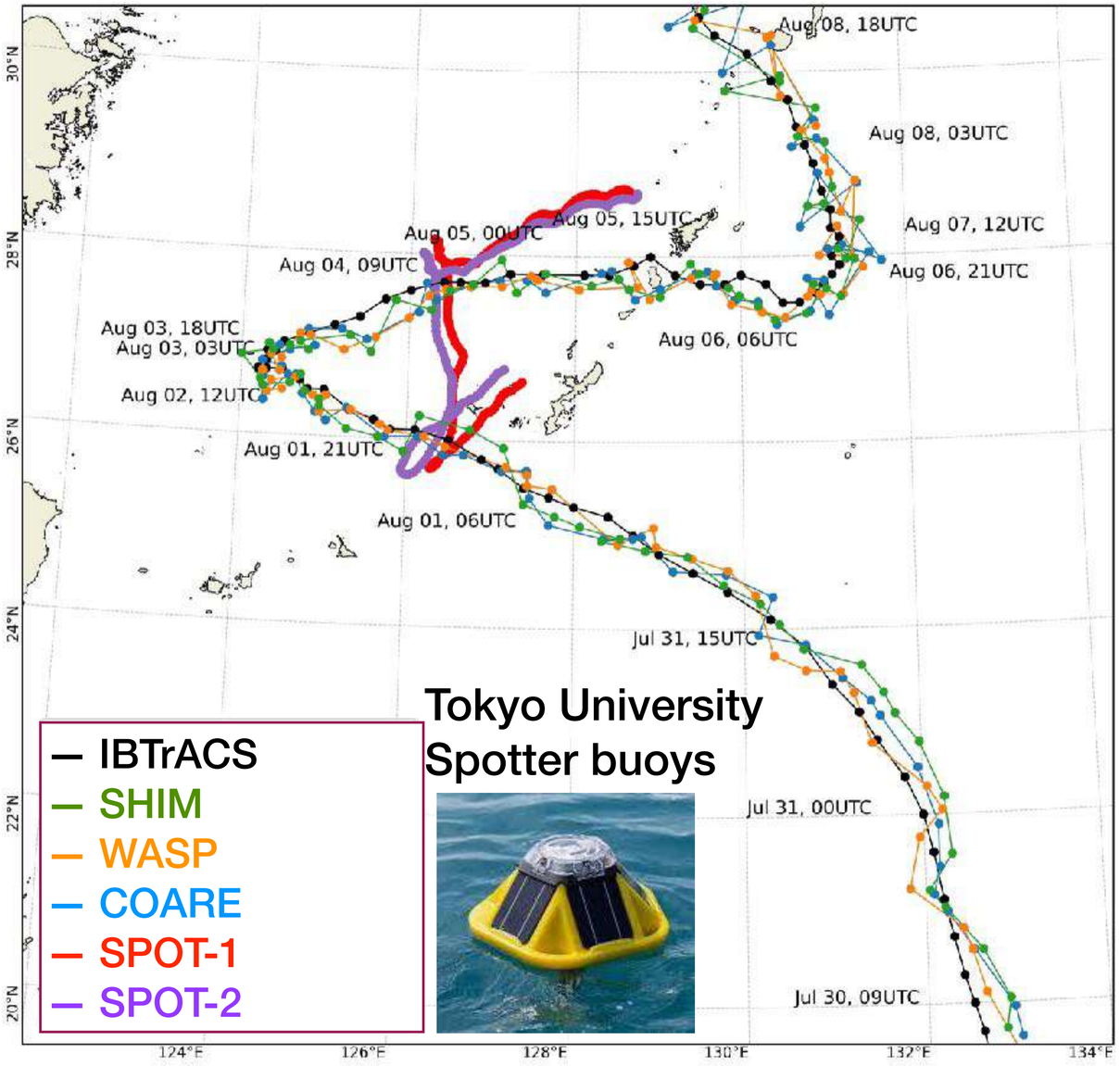


Seo et al. (2007, 2016); Sauvage et al. (2023, 2024)

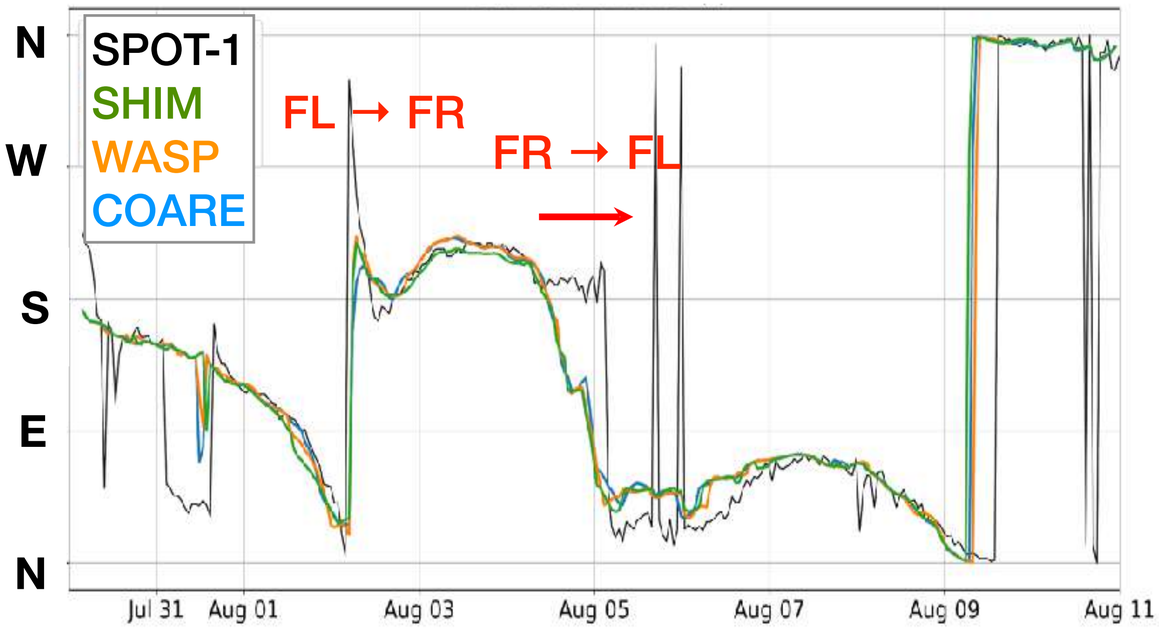
https://www.hyodaeseo.com/scoar/



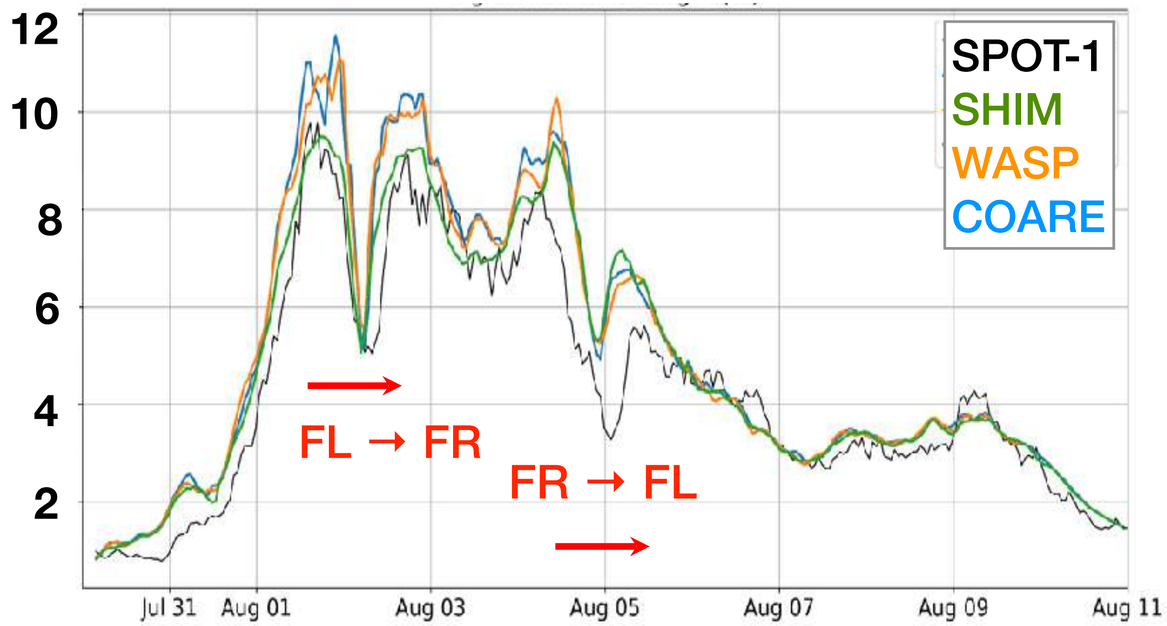
① Khanun's Sea State



Peak wave direction



Significant wave height



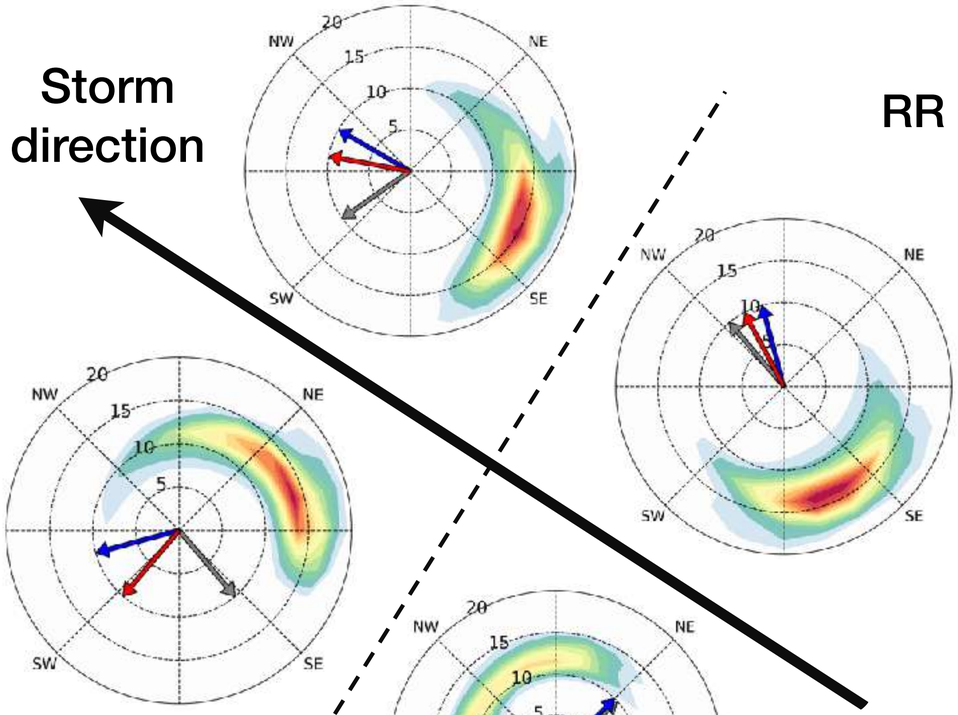
Directional Wave-Wind Coupling

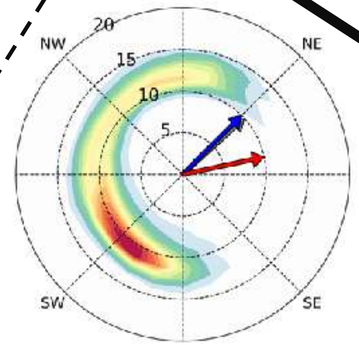
↖ mean wave direction
↖ peak wave direction
↖ wind direction

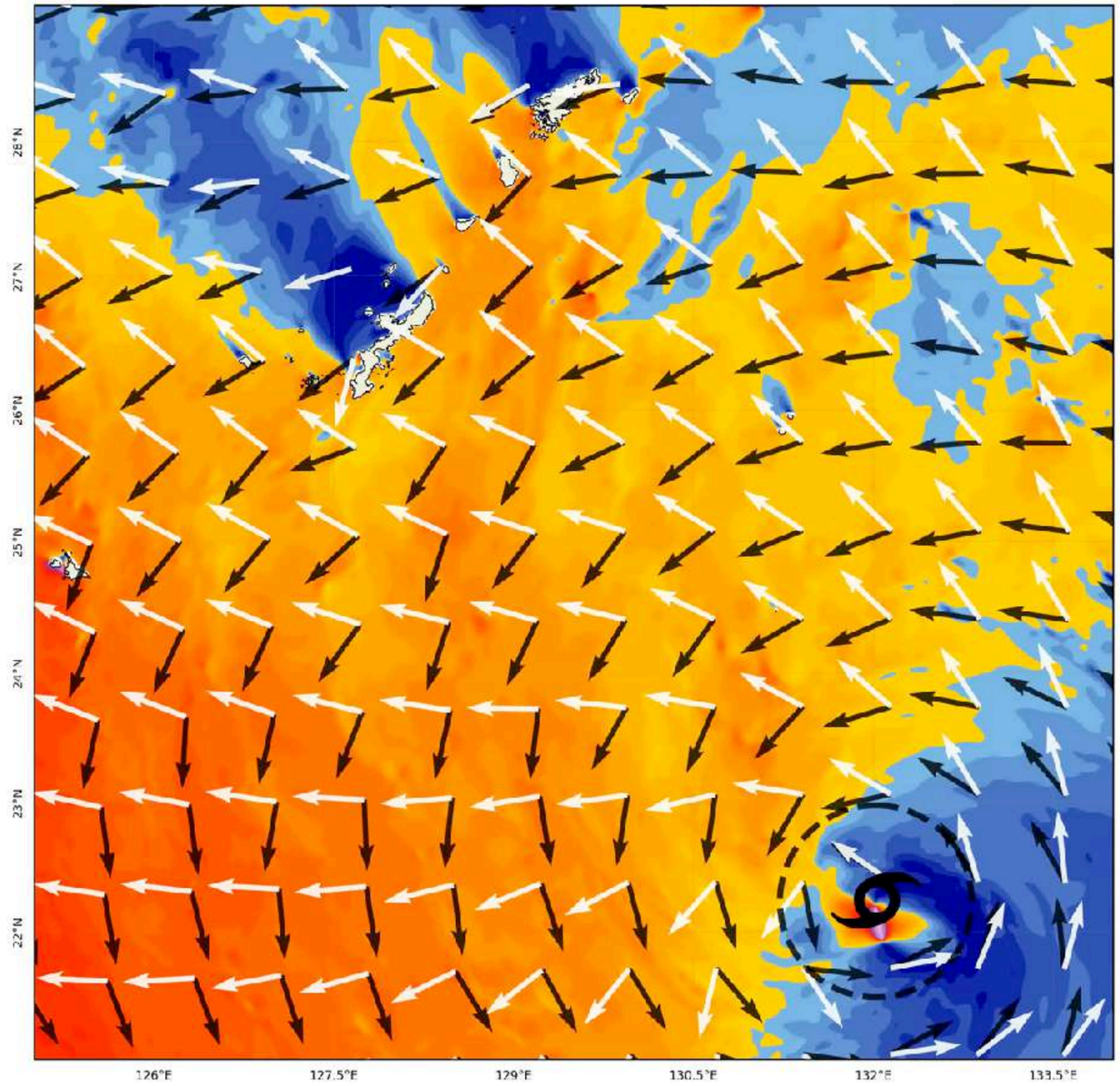
Storm direction

RR

FL







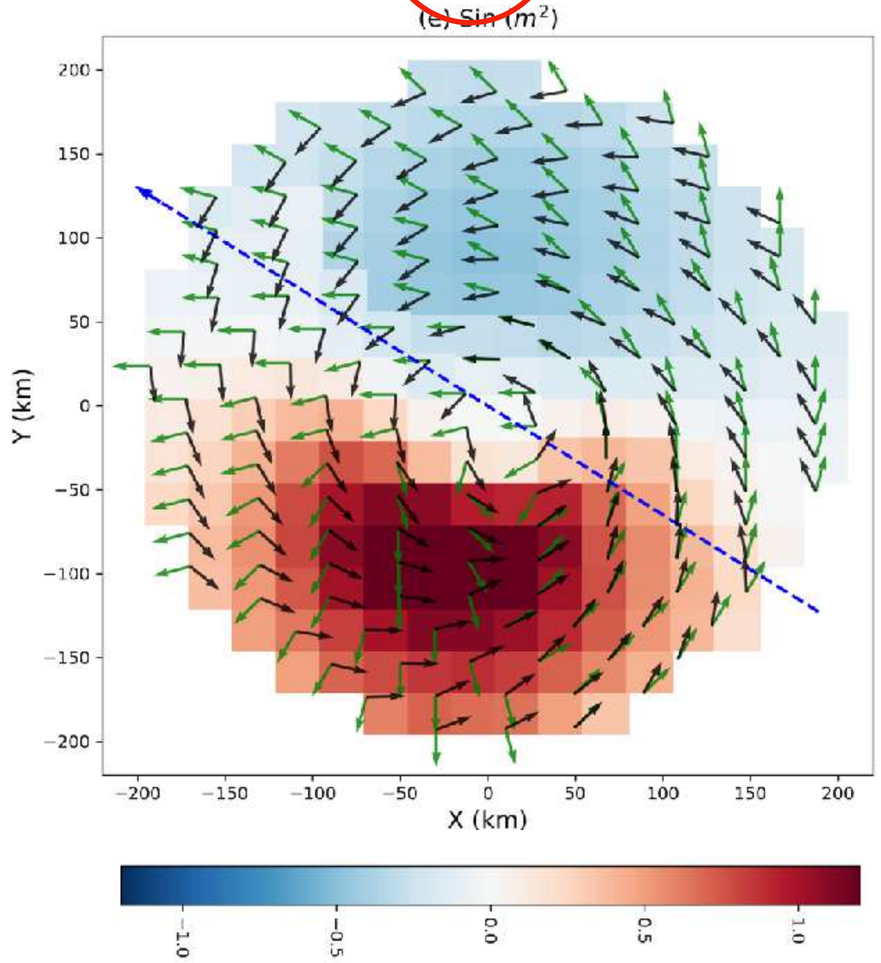
↖ mean wave direction
↖ wind direction

θ (°)
180
Opposed
135
90
Crossing
45
Aligned
0

July 31—Aug 2

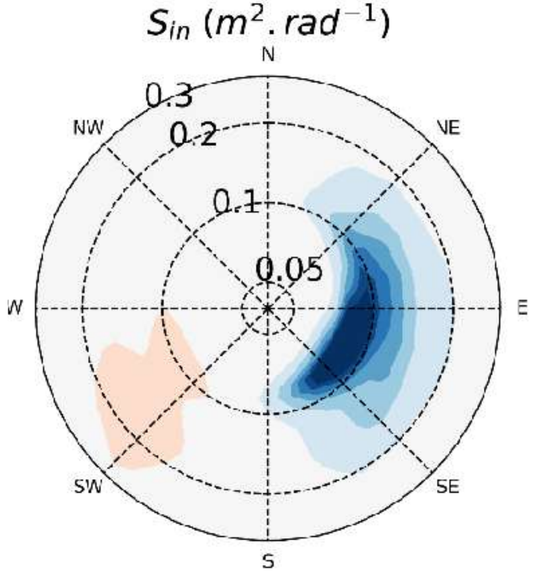
Spectral Source Term Diagnostics

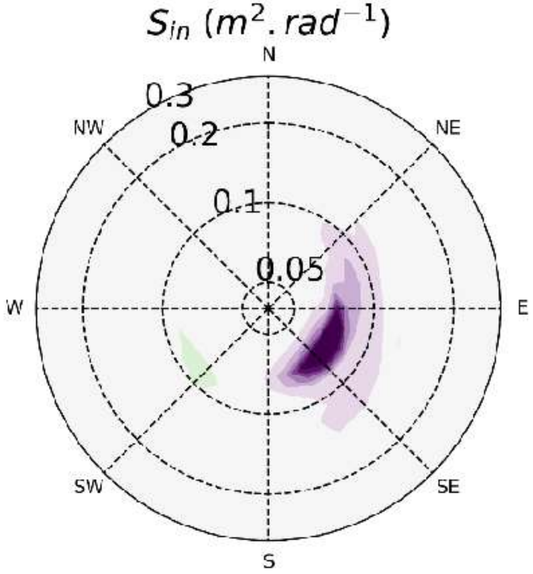
$$\frac{\partial N}{\partial t} + \nabla \cdot cN = \frac{S_{tot}}{\sigma} = \frac{S_{in}}{\sigma} + \frac{S_{ds}}{\sigma} + \frac{S_{nl}}{\sigma}$$



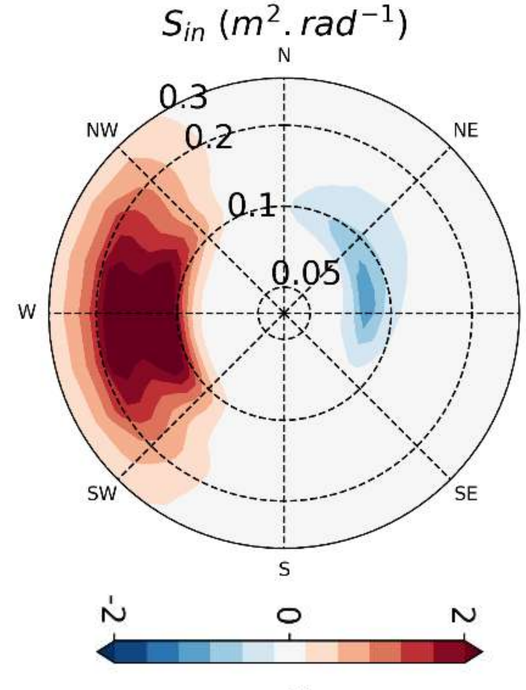
S_{in}: SHIM **S_{in}: SHIM-COARE**

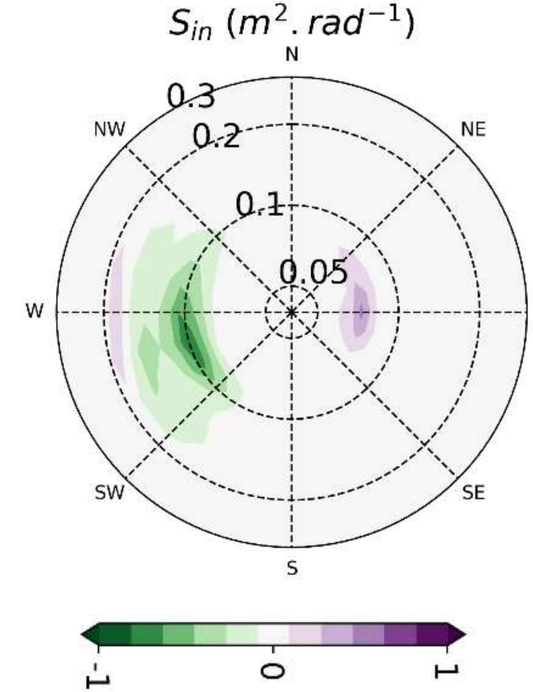
Right



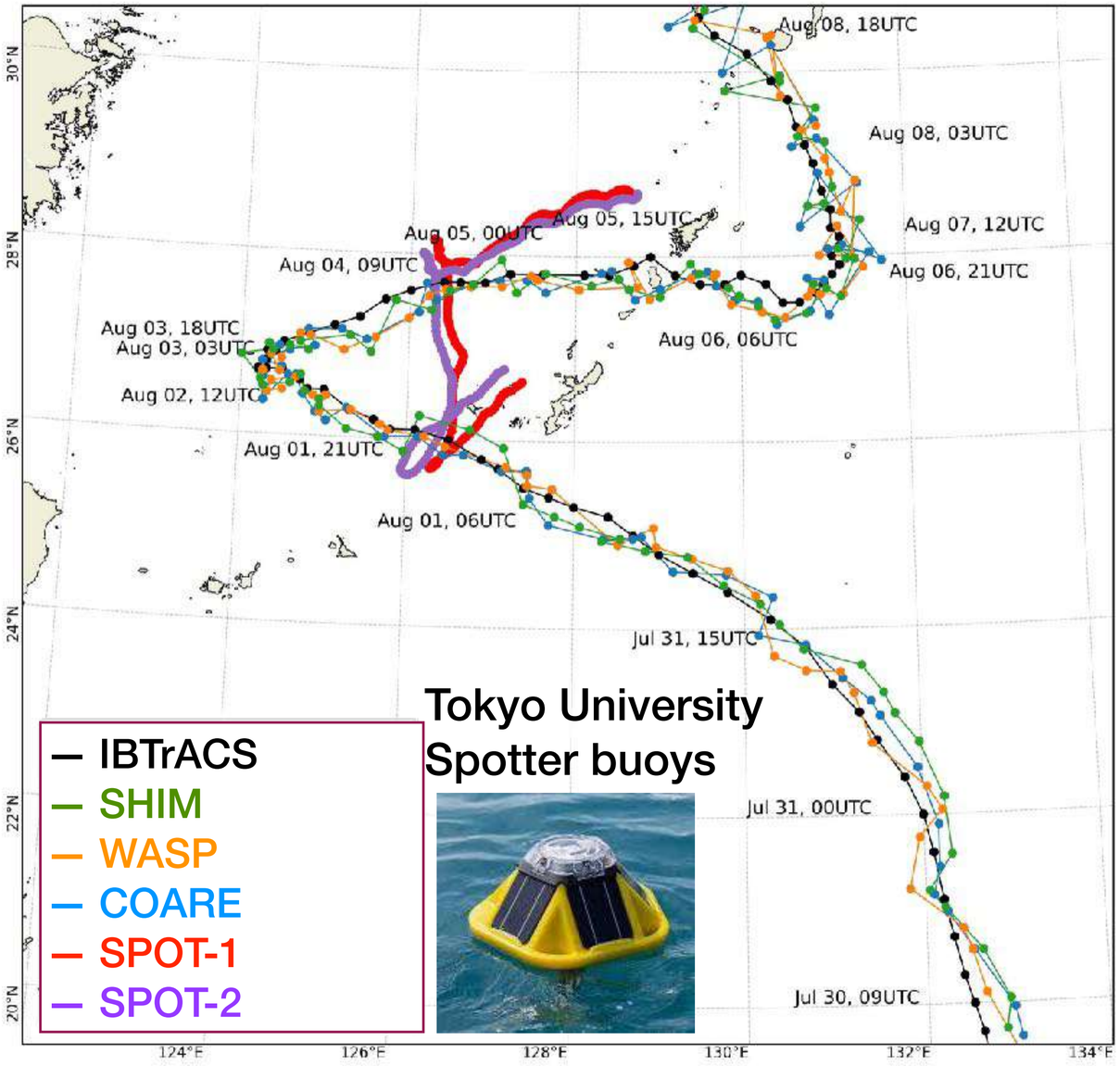


Left

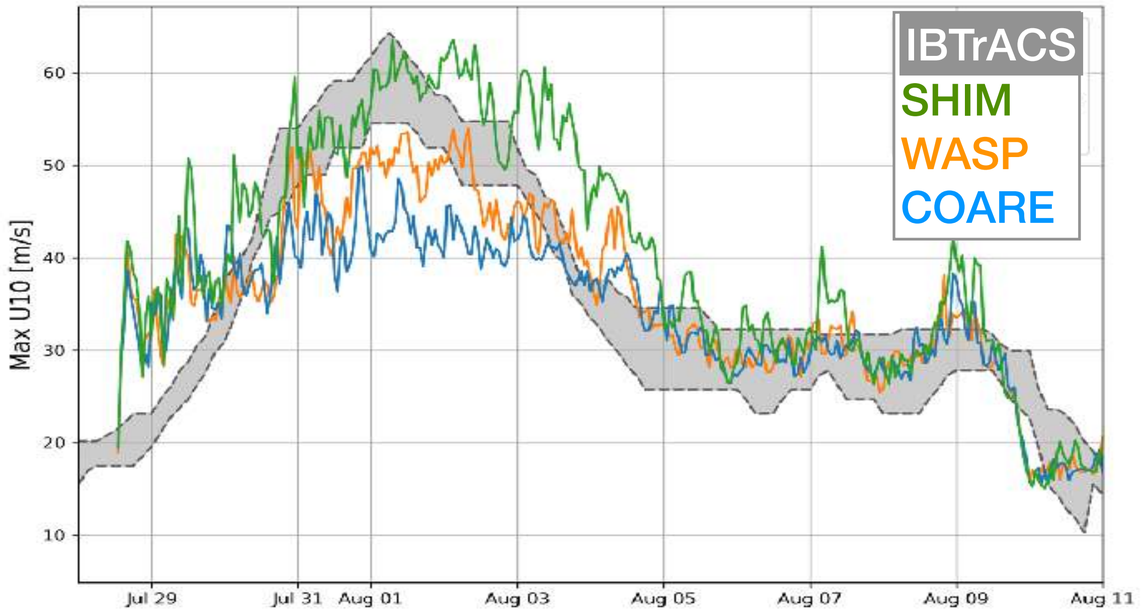




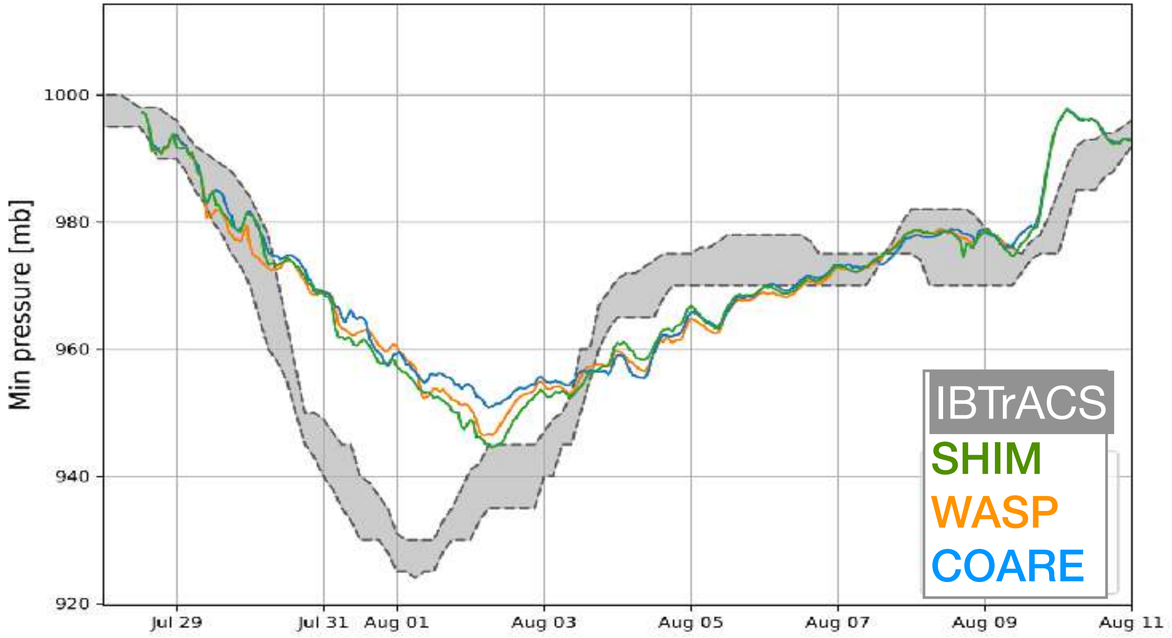
② Khanun's Intensity



Maximum wind speed (m/s)



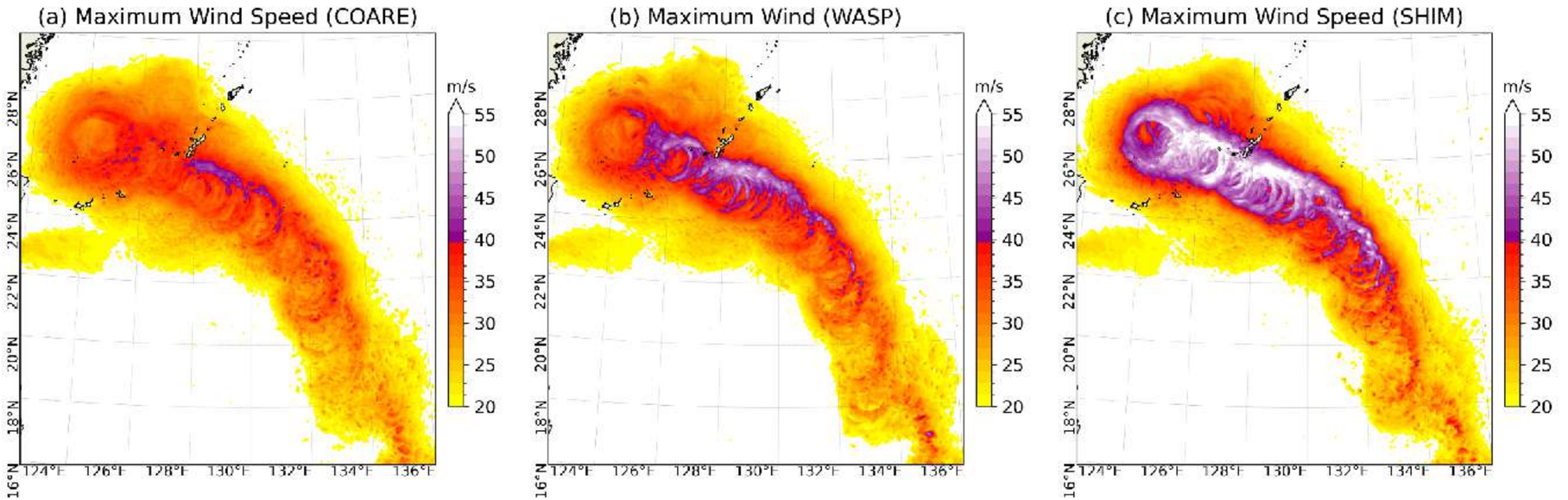
Minimum Sea level pressure



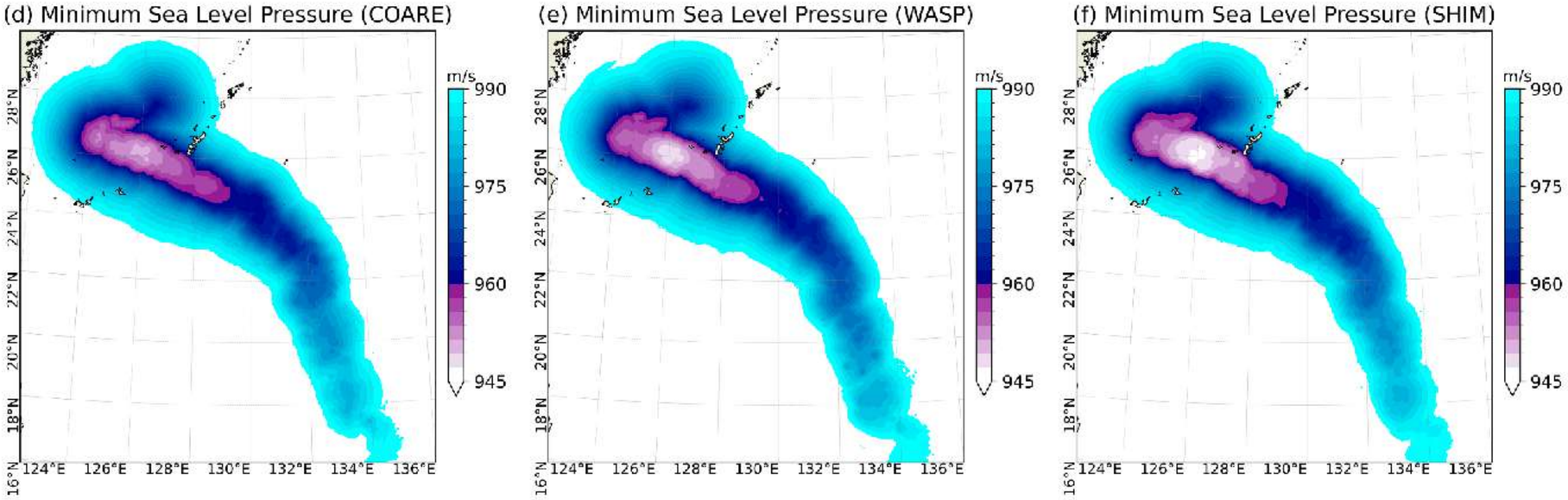
Simulated Storm Intensity

COARE WASP SHIM

V_{max}



SLP_{min}



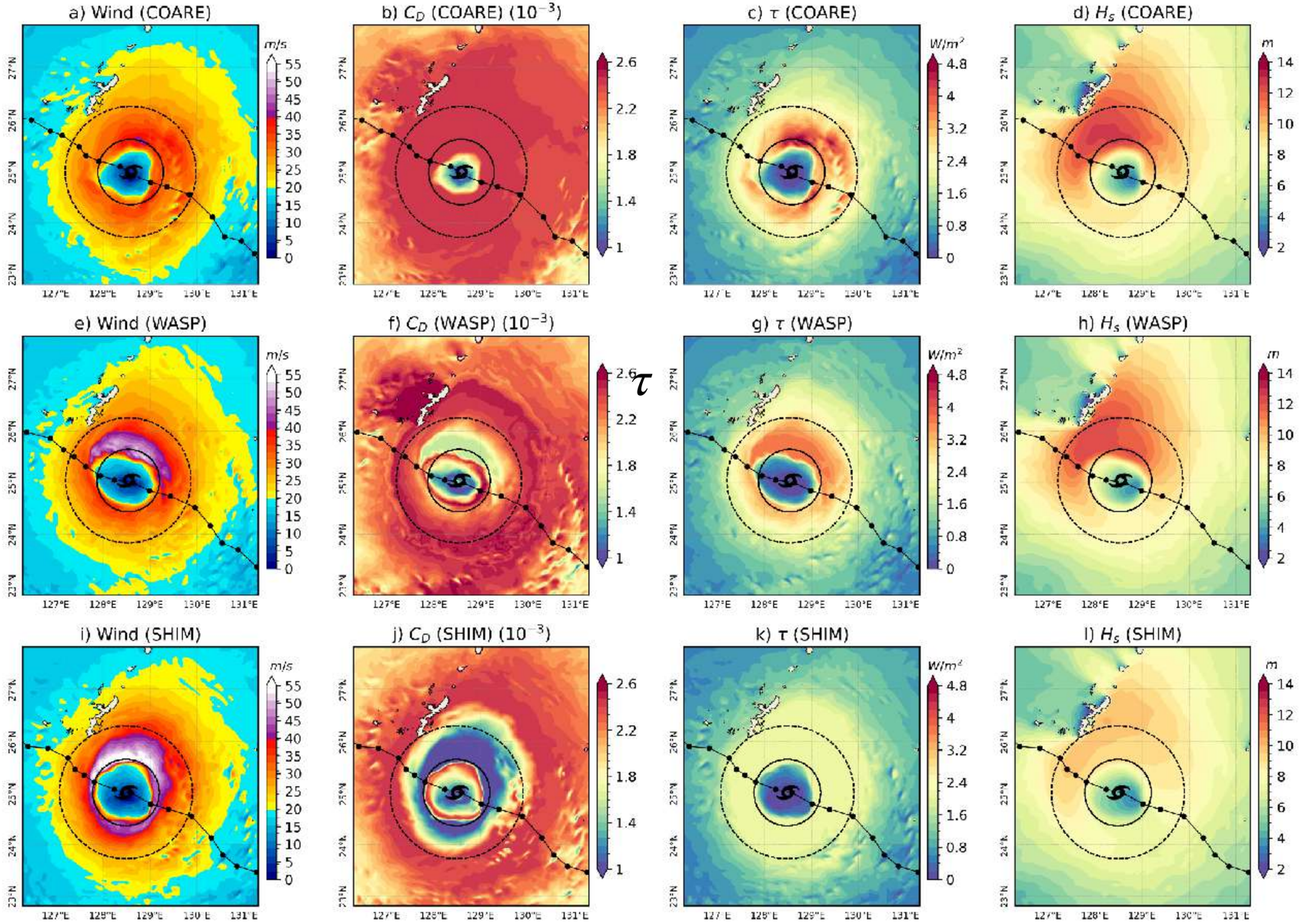
Drag Coefficients, Wave Energies, and Air-Sea Momentum Exchanges

U_{10} C_D τ H_s

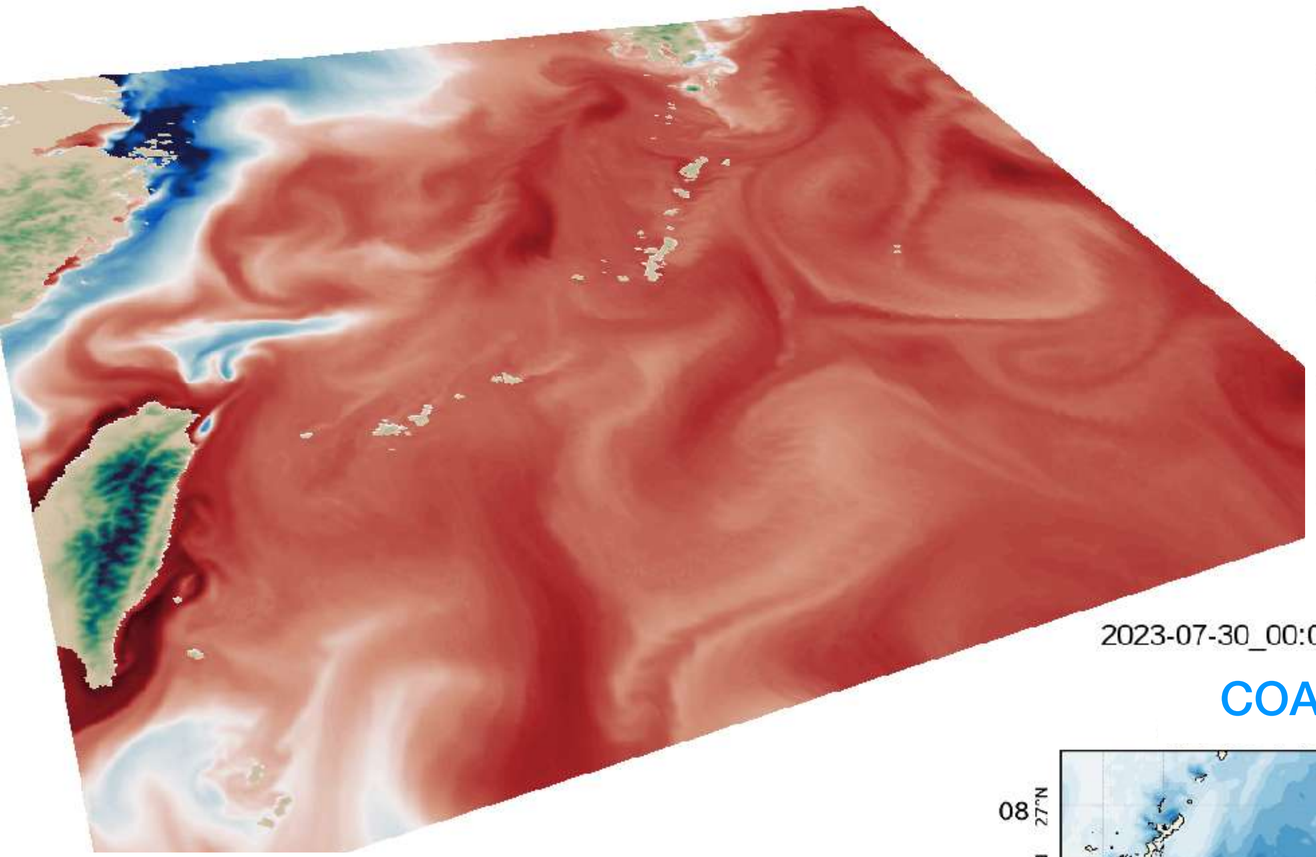
COARE

WASP

SHIM



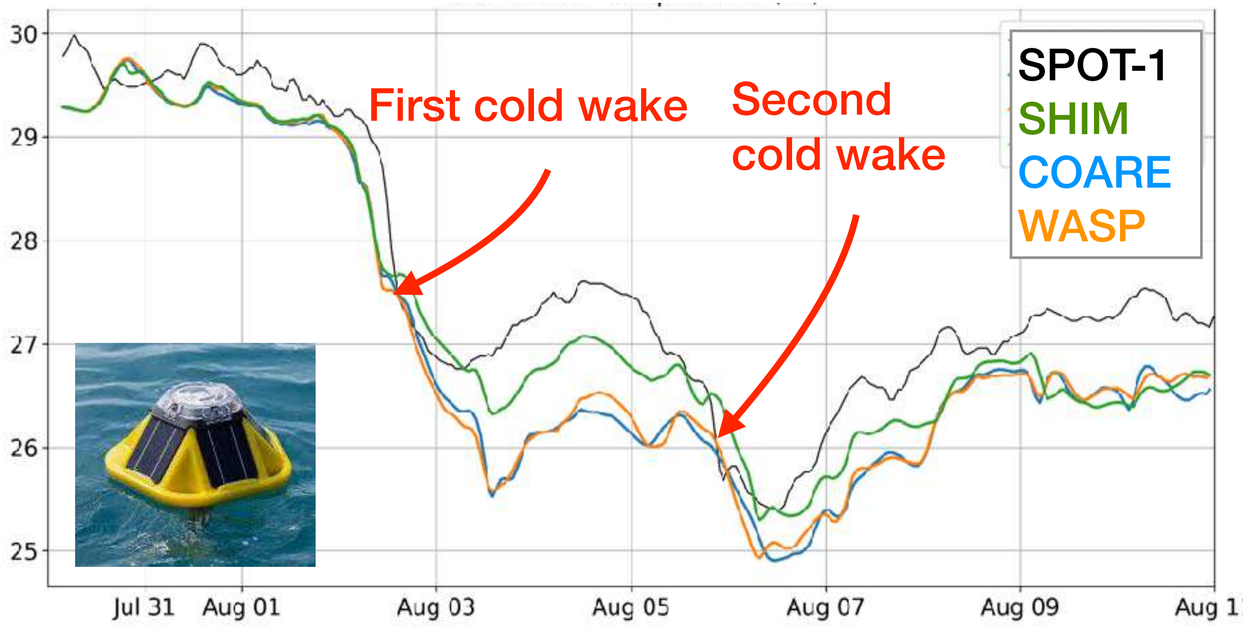
③ Khanun induced cold wake



Wind (m/s)
45.00
39.00
33.00
27.00
21.00
15.00

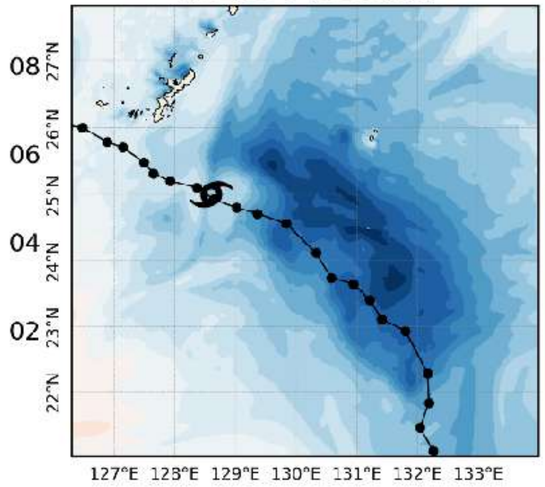
SST (C)
30.50
29.00
27.50
26.00
24.50
23.00

Sea surface temperature

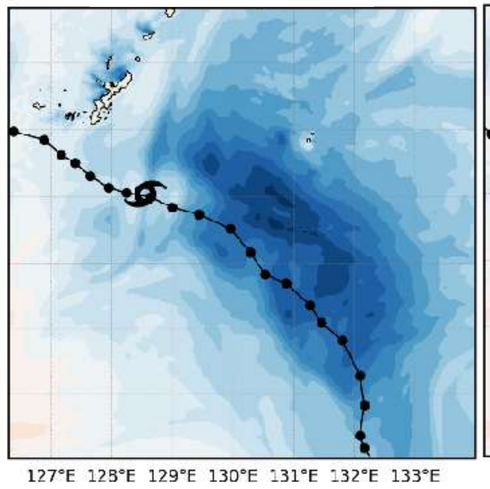


2023-07-30_00:00:00

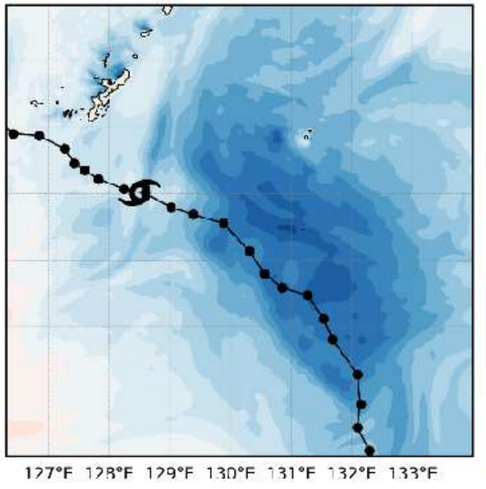
COARE

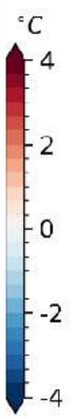


WASP



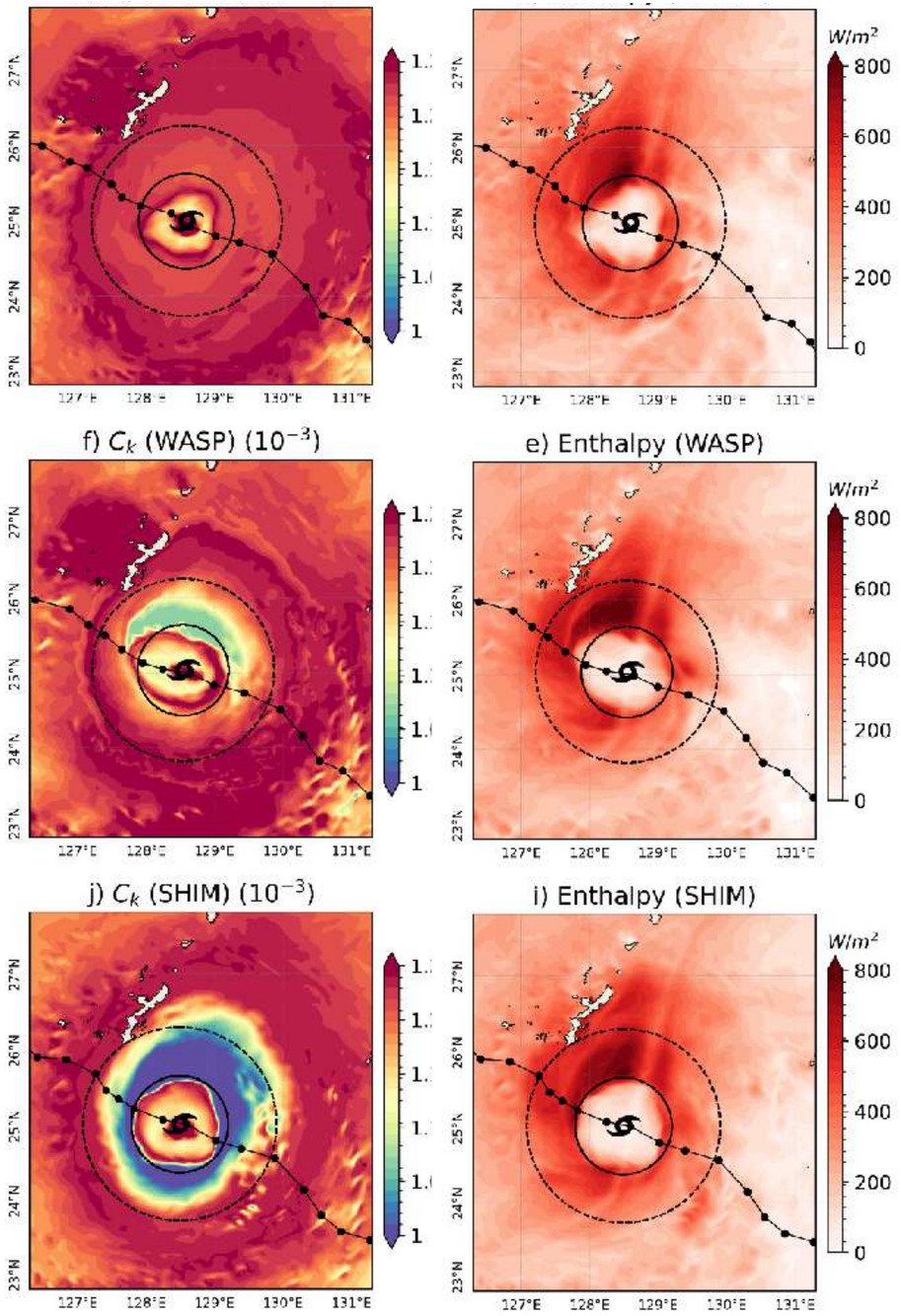
SHIM





Ck

Enthalpy flux



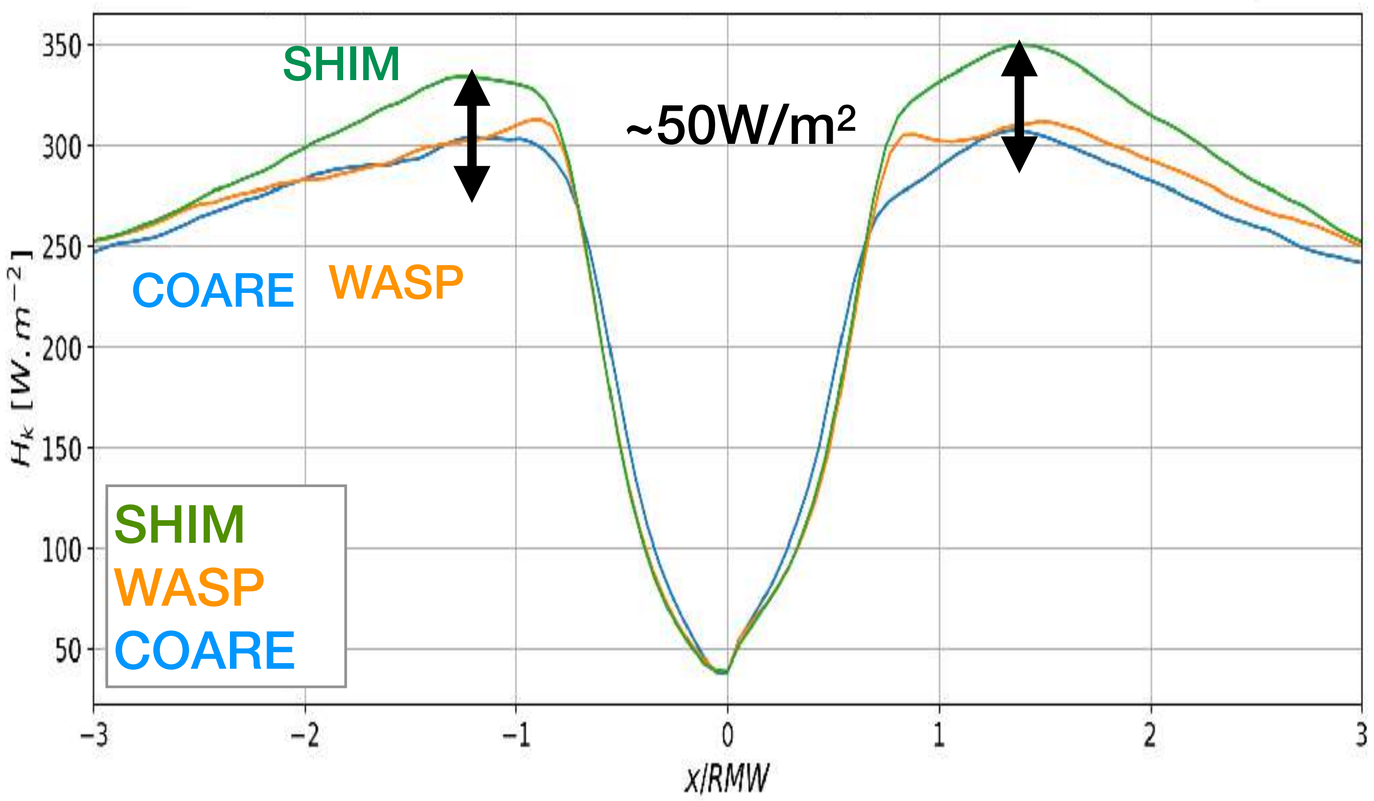
COARE

WASP

SHIM

Enthalpy flux: Feedback effect

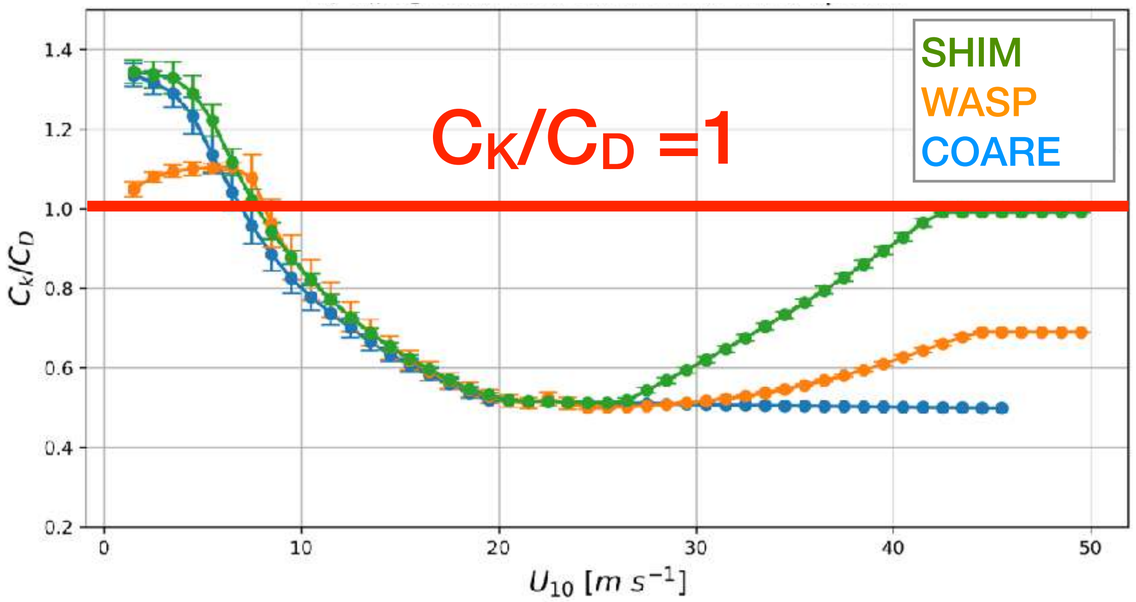
Time-averaged Enhlaphy flux

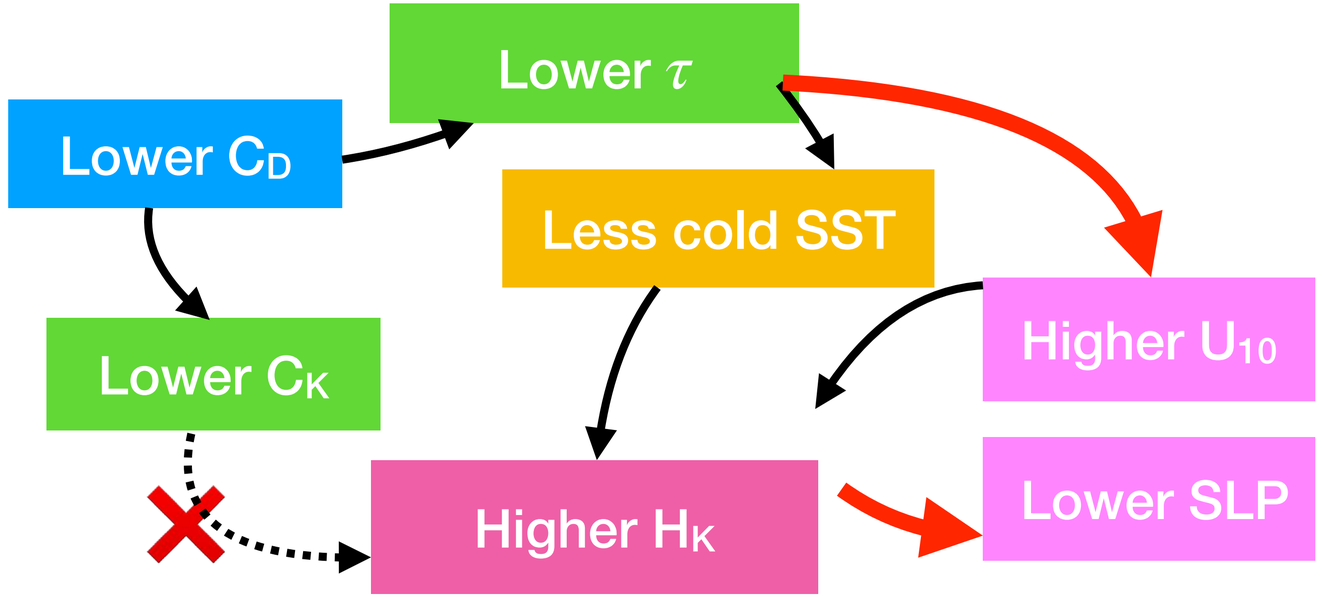


Summary and Discussion

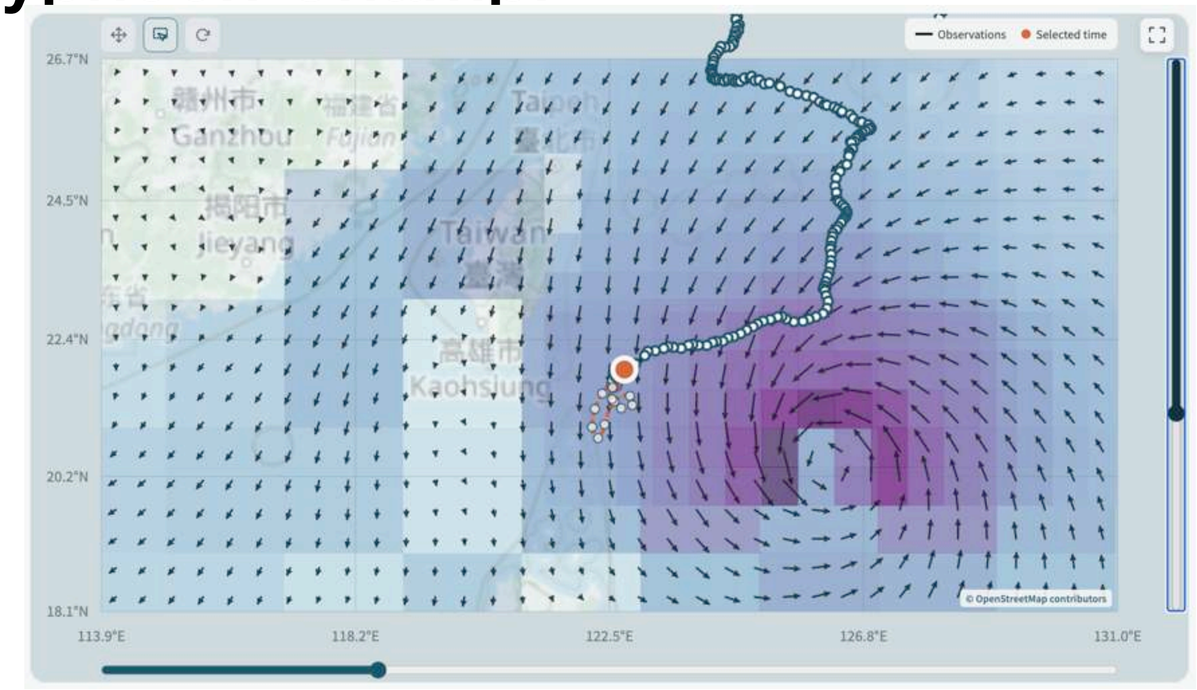
C_D behavior in high winds (>30 m/s) represents the strong spatial variability in sea states and wave-wind directional coupling, which feed back on storm intensity by modulating air-sea fluxes.

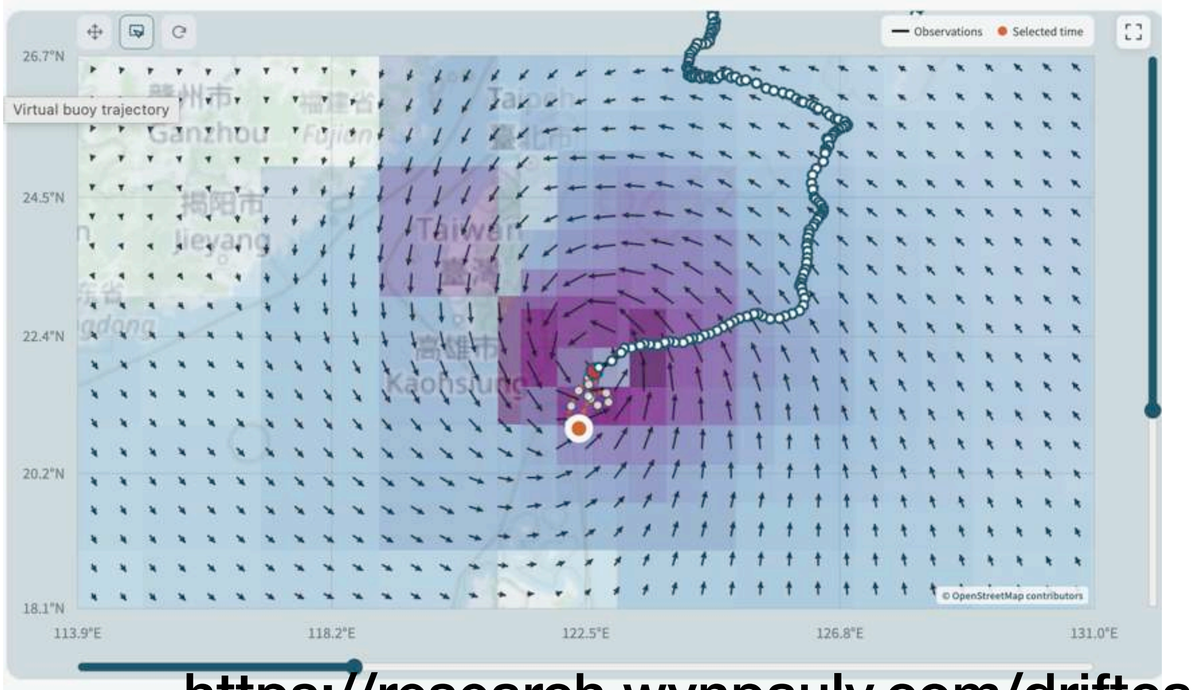
C_K/C_D ratio



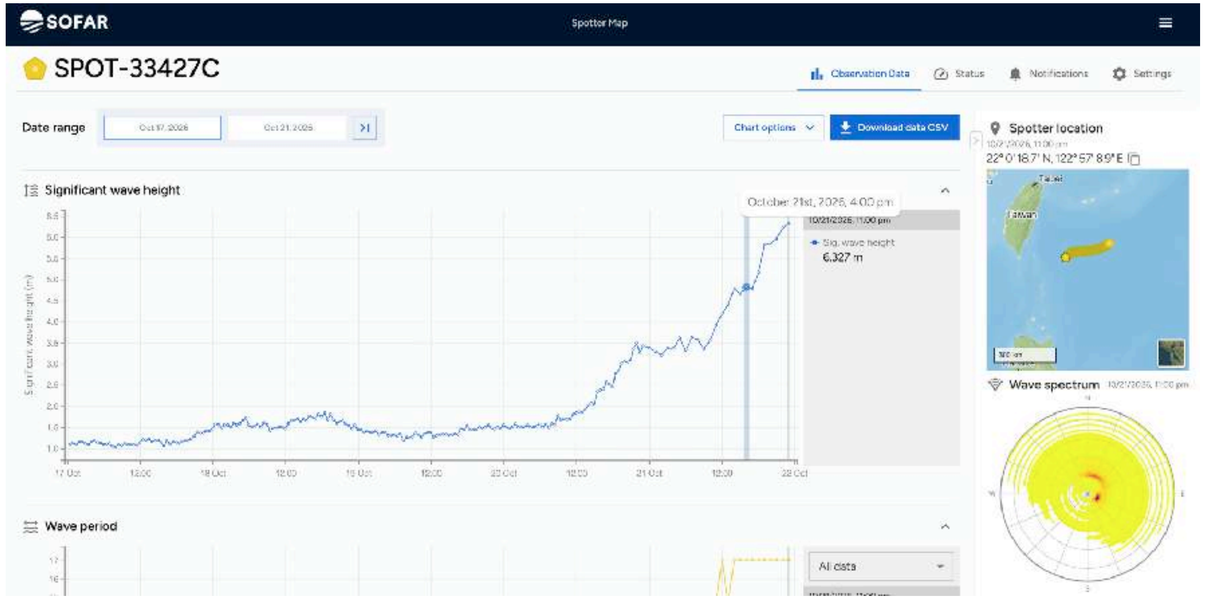


Typhoon Champi





https://research.wynpauly.com/driftcaste



Drag Coefficient and Storm Potential Intensity

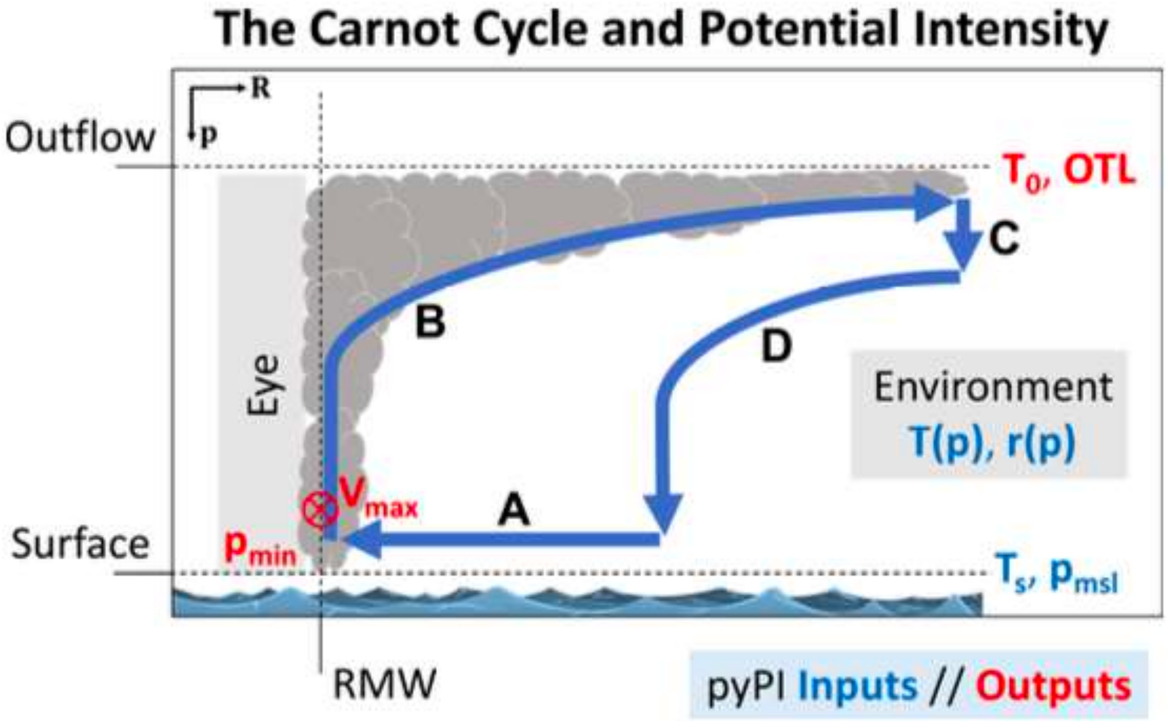
In the equilbirum condtion, TC intensity is contrained by the balance between energy added at the warm ocean surface and energy lost in the cool outflow region aloft (Emanuel 1988)

$$V_{max}^2 = \frac{(\overline{T} - T_o)}{T_o} \frac{C_K}{C_D} (K_{SST} - k)$$

TC outflow tempearture

pre-storm ocean temp.

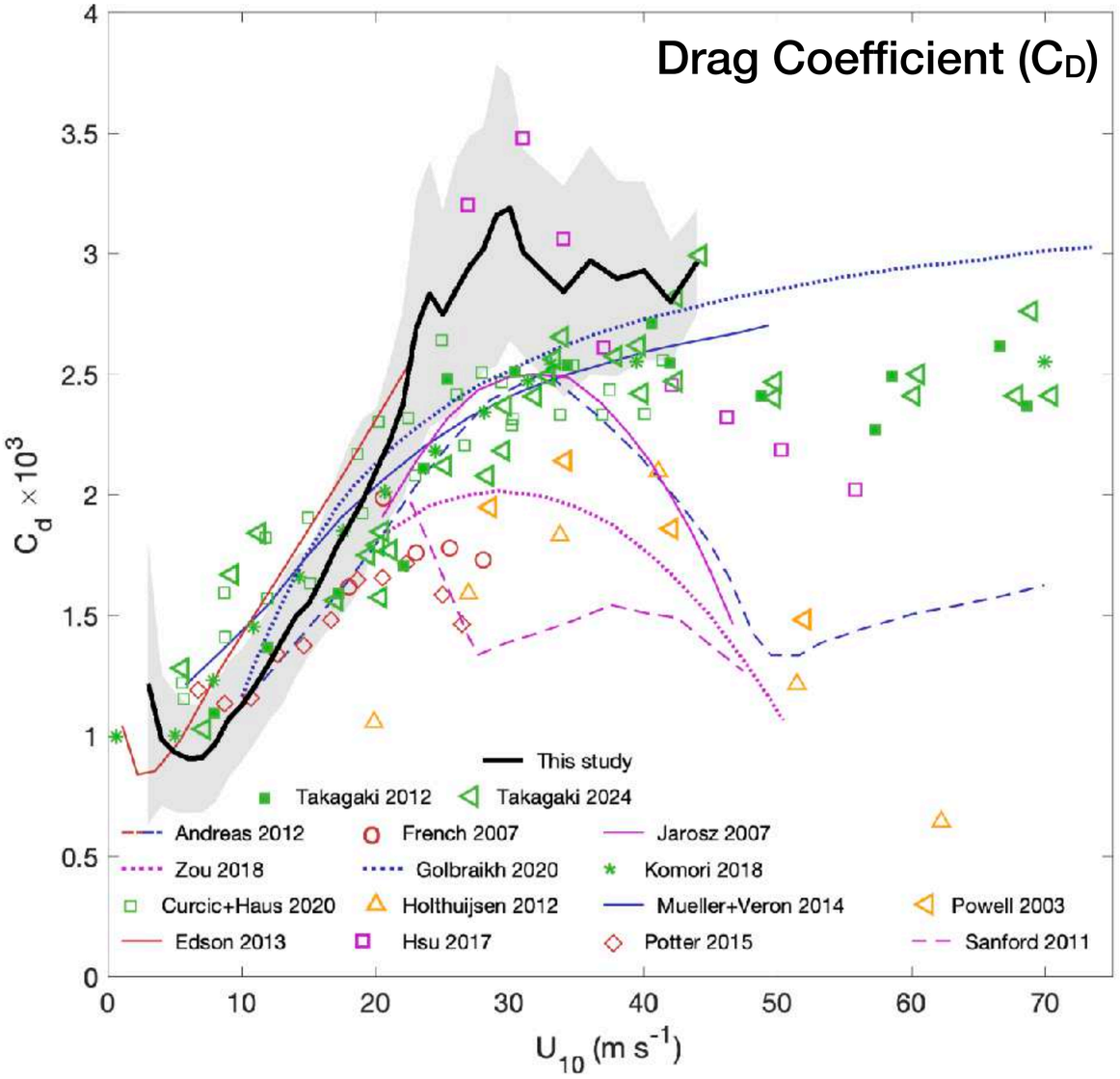
air-sea enthalpy difference



Gilford 2021

C_K, C_D, k, & K_{SST} all vary with sea state and its spatial variability under high wind.
Sea states and wave-supported roughness length are key parameters.

Sea state and drag coefficient under very high winds





https://www.youtube.com/results?search_query=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3DTzgdSSb0T98

The hurricane foam coverage and sea spray form a slippery surface: Wind energy input to waves ceases to increase while wind speeds increase

Foltz et al. 2026

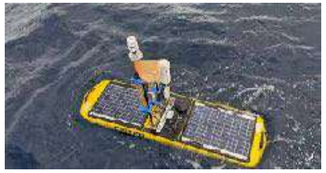
Typhoon Khanun (2023)

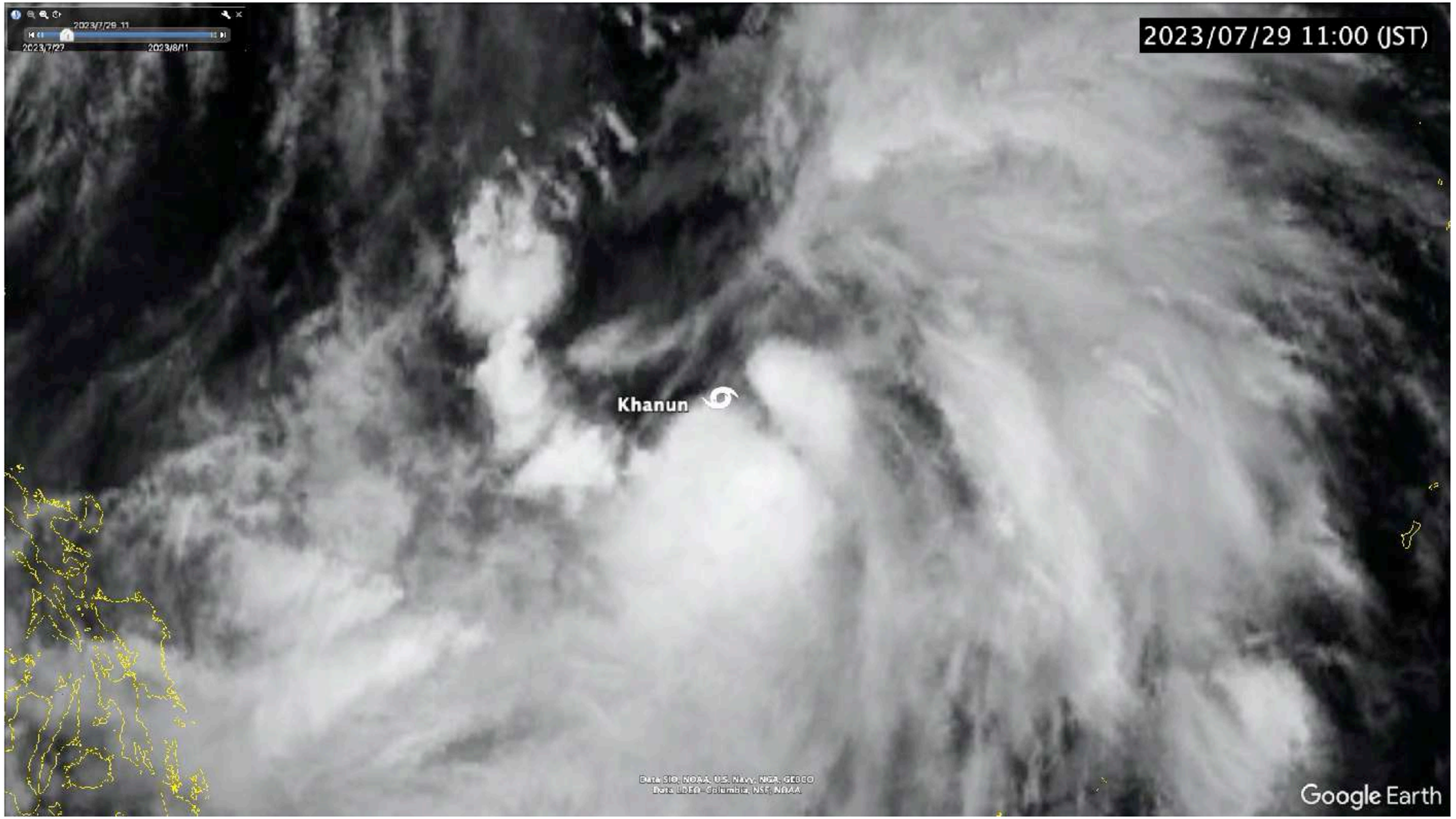
5x



(2 survived and neared the storm center twice)

2x





Courtesy: Prof. Satoshi Mitarai (OIST)